

TECHNICAL PROJECT REPORT

AI-Powered Job Recruitment & Candidate Management System

Built on Salesforce CRM with Agentforce AI Framework

Platform Environment: Salesforce Developer Edition Org (`orgfarm-ea3bff8dc7-dev-ed`)

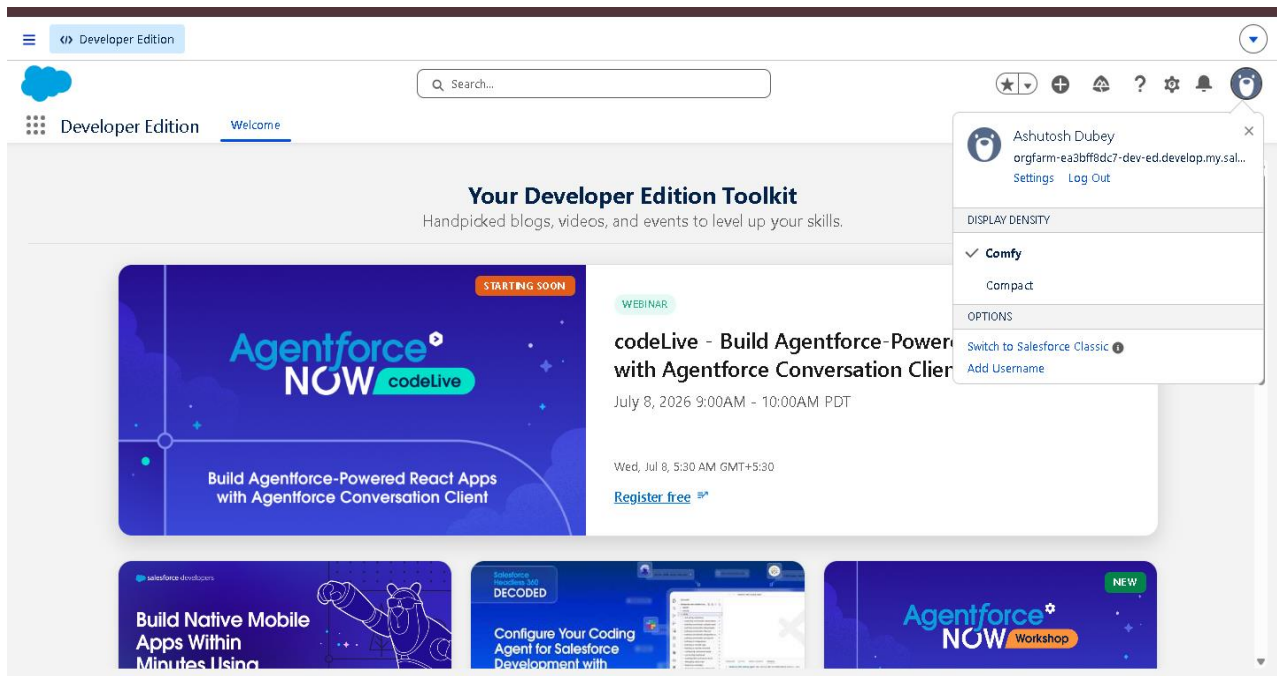


Figure 1.1: Salesforce Developer Edition active org validation for Ashutosh Dubey.

System Architect / Administrator: Ashutosh Dubey

Team ID: SWTID-2026-2645

Skill Wallet ID: SWUID2026234515

Email ID: ashutoshdubey231001@gmail.com

Academic / Internship Track:

SmartBridge Virtual Internship Program

Salesforce Certified Administrator with AI Agentforce Specialization

Date of Submission: July 15, 2026

Certificate of Completion

To Whom It May Concern

This is to certify that the project titled "**AI-Powered Job Recruitment & Candidate Management System**" has been successfully configured and executed by **Ashutosh Dubey** under the SmartBridge Virtual Internship Program - Salesforce Certified Administrator with AI Agentforce Specialization.

The system metadata, data schemas, automated record-triggered processing frameworks, validation rules, and generative AI models documented within this technical report represent a fully functional, deployment-ready platform solution. All development work detailed herein was conducted natively within a zero-custom-Apex Salesforce configuration layer, adhering directly to enterprise standards for platform cloud scalability, data integrity governance, and role-based access controls.

SmartBridge Program Coordinator

Technical Mentorship Directorate

Acknowledgement

I would like to express my sincere technical gratitude to the core engineering mentors and evaluation coordinators of the SmartBridge Virtual Internship Program. The structured learning tracks covering advanced Salesforce administration, declarative process optimization via Flow Builder, and enterprise generative AI deployment via the Agentforce framework provided the professional foundations required to configure this system architecture.

I also extend acknowledgment to the global Salesforce Trailhead developer network for providing reference models concerning relational data protection, cross-object field mapping, and secure profile validation parameters. Finally, I thank my peer teams for their collaborative feedback during the programmatic testing and simulation loops of this system build.

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1. Executive Summary & Business Rationale

Modern enterprise organizations process hundreds of candidate applications daily during scaling periods. Traditional talent acquisition workflows that rely heavily on manual tracking systems, standalone spreadsheets, and unlinked email strings create severe operational challenges, including:

- **Data Redundancy:** Inability to trace duplicate candidate records across concurrent recruitment cycles.
- **Application Attrition:** High risk of dropping qualified applicant profiles due to manual routing flaws.
- **Recruitment Latency:** Delayed interview scheduling caused by unstructured coordination loops.
- **Subjective Inconsistency:** Screenings lack standardized evaluation criteria, introducing bias.
- **Operational Opacity:** Management tiers lack visibility into live applicant pipelines and aggregate team productivity.

To eliminate these operational cracks, this project establishes a centralized, Salesforce CRM-driven recruitment application. This system integrates tracking structures, automates sequential workflows, executes real-time AI-based evaluations, accelerates managerial approval chains, and optimizes recruiter velocity without introducing custom programmatic overhead.

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2. Project Scope & Functional Lifecycle

The implementation scope covers the deployment of a scalable recruitment platform built exclusively via declarative tools, advanced automations, and Agentforce AI within the Salesforce Lightning ecosystem. Programmatic components (Apex, Visualforce, LWC) are omitted to minimize technical debt and ensure long-term ease of maintenance. The application drives and tracks the complete, end-to-end recruitment lifecycle through five sequential operational phases:

Candidate → Screening → Interview → Offer → Hiring

Core development steps are split across five distinct engineering phases: Requirement Analysis and Planning; Backend Schema Configuration; UI/UX Customization; Data Migration & Deduplication Security; and Deployment Validation with Change Management.

3. Relational Data Model & Schema Architecture

The structural data layer is constructed using highly optimized, custom relational schemas mapped directly to native Salesforce object structures. The architecture focuses on two core custom entities built within the Object Manager layer:

- **Job Opening (Job_Opening__c):** Represents a single open job position. It tracks corporate structural metrics such as department parameters, target vacancy counts, job

status tracking picklists, and baseline financial allocations. The record name field is configured as **Job Title** (Data Type: Text).

The screenshot shows the Salesforce Setup page for creating a new custom object. The page is titled "New Custom Object" and shows the "Custom Object Definition Edit" section. The "Custom Object Information" section is visible, with fields for Label, Plural Label, Object Name, and Description. The Label is "Job Opening", Plural Label is "Job Openings", and Object Name is "Job_Opening". The Description field is empty. The "Context-Sensitive Help Setting" is set to "Open the standard Salesforce.com Help & Training window". The "Content Name" is set to "None".

The screenshot shows the Salesforce Setup page for creating a new custom object. The page is titled "New Custom Object" and shows the "Custom Object Definition Edit" section. The "Custom Object Information" section is visible, with fields for Record Name, Data Type, and Optional Features. The Record Name is "Job Title", Data Type is "Text", and Optional Features are "Allow Reports", "Allow Activities", "Track Field History", "Allow in Chatter Groups", and "Enable Licensing". The "Object Classification" section is visible, with fields for Allow Sharing, Allow Bulk API Access, and Allow Streaming API Access. The "Deployment Status" section is visible, with fields for In Development and Deployed. The "Search Status" section is visible.

“Figure 3.1: Step-by-step creation of Custom Object Job Opening with history tracking enabled.”

Candidate Application (Candidate_Application__c): A custom junction object configured to manage the recruitment lifecycle. To prevent orphaned transactional entries, the record name field uses an automated sequencing format set to **Application ID** (Data Type: Auto Number, format: CA- {0000}, starting with number 1).

Custom Object Definition Edit

Custom Object Information

The singular and plural labels are used in tabs, page layouts, and reports.

Label: Example: Account

Plural Label: Example: Accounts

Starts with vowel sound: ☐

The Object Name is used when referencing the object via the API.

Object Name: Example: Account

Description:

Context-Sensitive Help Setting: ☒ Open the standard Salesforce.com Help & Training window
☐ Open a window using a Visualforce page

Content Name:

Enter Record Name Label and Format

Enter Record Name Label and Format

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name field is always called "Name" when referenced via the API.

Record Name: Example: Account Name

Data Type: Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.

Display Format: Example: A-0000 What is This?

Starting Number:

Optional Features

- ☒ Allow Reports
- ☒ Allow Activities
- ☒ Track Field History
- ☐ Allow in Chatter Groups
- ☐ Enable Licensing

Object Classification

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more.](#)

- ☒ Allow Sharing
- ☒ Allow Bulk API Access
- ☒ Allow Streaming API Access

“Figure 3.2: Custom junction object Candidate Application setup with Auto-Number formatting.”

To ensure high relational integrity, the data model maps parent-child object connections using dual ****Master-Detail Relationships**** from the `Candidate_Application__c` child record. It links directly up to the `Job_Opening__c` table and the standard `Contact` (Candidate) object ledger. If a master candidate profile or parent job opening record is deleted, the platform cascade-deletes related candidate applications automatically, keeping data clean and avoiding broken references.

4. Custom Fields & Relationships Directory

The tables below outline the structural custom field allocations configured across objects within the Object Manager layer of the developer environment:

4.1 Standard Contact Object (Candidate Ledger Extensions)

Field Label	Field API Name	Data Type	Configuration Parameters / Picklist Values
Candidate ID	Candidate_ID__c	Auto Number	Display Format: C-{0000} (Starting Number: 1)
Experience Level	Experience_Level__c	Picklist	Fresher, Mid-Level, Senior
Application Status	Application_Status__c	Picklist	New, Screening, Interview, Offer, Hired

Contact | Salesforce

orgfarm-ea3bff8dc7-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/Contact/FieldsAndRelationships/new

Developer Edition

Setup Home Object Manager

SETUP > OBJECT MANAGER

Contact

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Hierarchy Columns

Step 2. Enter the details

Step 2 of 4

Field LabelExperience Level

Values

Use global picklist value set

Enter values, with each value separated by a new line

Fresher
Mid Level
Senior

Display values alphabetically, not in the order entered

Use first value as default value

☒ Restrict picklist to the values defined in the value set

Field NameExperience_Level

Description

Contact | Salesforce

orgfarm-ea3bff8dc7-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/Contact/FieldsAndRelationships/page?address=%2F00Nfj000041Tj0%2Fe%3...

Developer Edition

Setup Home Object Manager

SETUP > OBJECT MANAGER

Contact

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Hierarchy Columns

Scoping Rules

Field LabelCandidate ID

Field NameCandidate_ID

Description

Help Text

Data OwnerUser

Field Usage--None--

Data Sensitivity Level--None--

Compliance Categorization

Available

PII
HIPAA
GDPR
PCI

Chosen

General Options

External ID☐ Set this field as the unique record identifier from an external system

Auto Number Options

Auto Number Display FormatC-0000

Change Field Type Save Cancel

Figure 4.1 displays two screenshots of the Salesforce Developer Edition interface, illustrating custom field parameters and synchronization values on the Contact layer.

The top screenshot shows the "Edit Contact Custom Field" page for the "Application Status" field. The field is defined with the following parameters:

- Field Label: Application Status
- Field Name: Application_Status
- Description: (Empty)
- Help Text: (Empty)
- Data Owner: User
- Field Usage: --None--
- Data Sensitivity Level: --None--
- Compliance Categorization: Available (PII, HIPAA, GDPR, PCI) / Chosen (Empty)
- General Options: Required (unchecked), Always require a value in this field in order to save a record (unchecked)

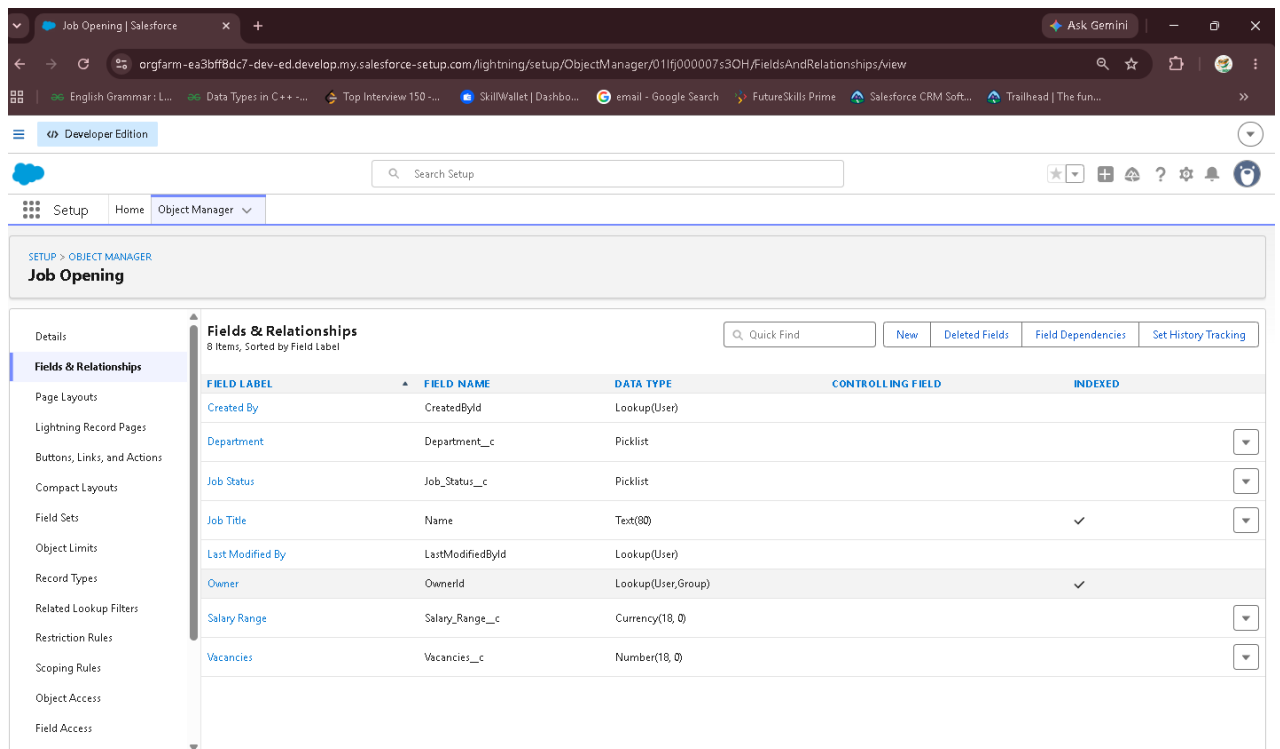
The bottom screenshot shows the "Fields & Relationships" page for the Contact object, displaying a list of fields and their synchronization values. The fields are sorted by Field Label.

Field Label	Field Name	Data Type	Synchronization Value
Department	Department	Text(80)	
Description	Description	Long Text Area(32000)	
Do Not Call	DoNotCall	Checkbox	
Email	Email	Email	✓
Email Opt Out	HasOptedOutOfEmail	Checkbox	
Experience Level	Experience_Level_c	Picklist	
Fax	Fax	Phone	
Fax Opt Out	HasOptedOutOfFax	Checkbox	
Gender Identity	GenderIdentity	Picklist	
Home Phone	HomePhone	Phone	

“Figure 4.1: Custom field parameters and synchronization values on the Contact layer.”

4.2 Custom Job Opening Object (Job_Opening__c)

Field Label	Field API Name	Data Type	Configuration Parameters / Picklist Values
Department	Department__c	Picklist	IT, HR, Finance, Sales
Salary Range	Salary_Range__c	Currency	Standard Currency Formatting (18, 2)
Vacancies	Vacancies__c	Number	Integer Formatting (Scale: 0; negative check active)
Job Status	Job_Status__c	Picklist	Open, Closed, On Hold



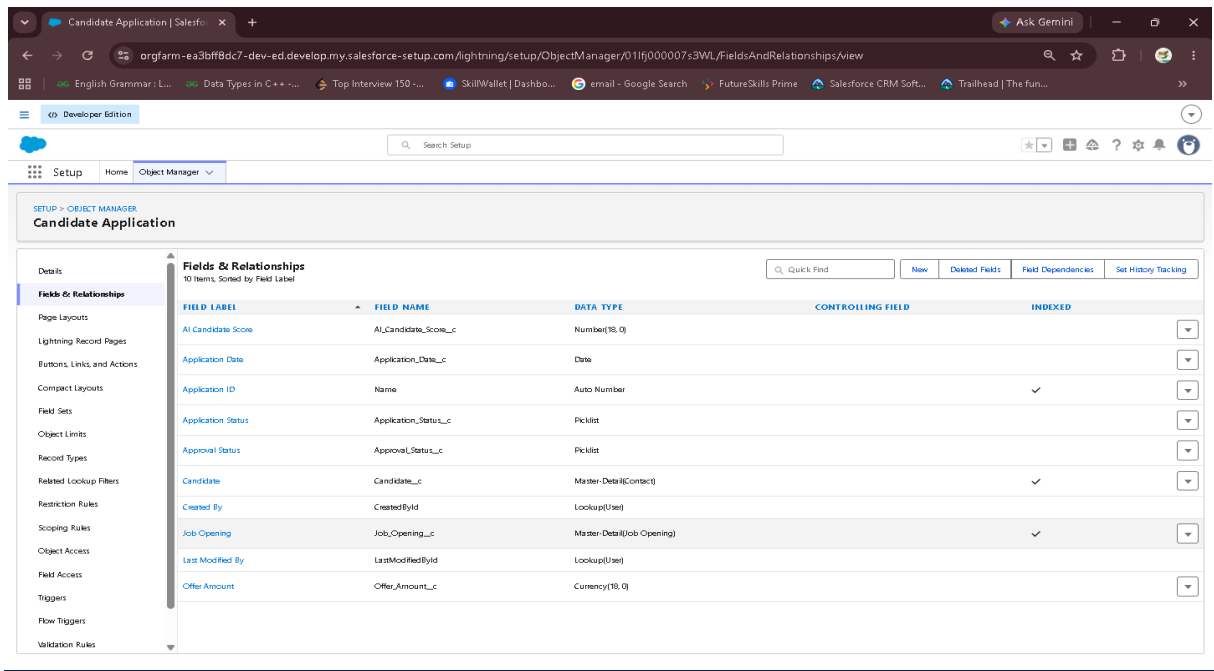
The screenshot displays the Salesforce Setup interface for the 'Job Opening' custom object. The 'Fields & Relationships' section is active, showing a list of 8 fields sorted by Field Label. The fields are: Created By, Department, Job Status, Job Title, Last Modified By, Owner, Salary Range, and Vacancies. Each field entry includes its Field Label, Field Name, Data Type, Controlling Field, and Indexed status. The interface also includes a search bar, navigation tabs (New, Deleted Fields, Field Dependencies, Set History Tracking), and a sidebar with various setup options like Page Layouts, Lightning Record Pages, and Object Limits.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Department	Department__c	Picklist		
Job Status	Job_Status__c	Picklist		
Job Title	Name	Text(80)		✓
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Salary Range	Salary_Range__c	Currency(18, 0)		
Vacancies	Vacancies__c	Number(18, 0)		

“Figure 4.2: Fields & Relationships directory matrix for Job Opening custom object.”

4.3 Custom Candidate Application Object (Candidate_Application__c)

Field Label	Field API Name	Data Type	Configuration Parameters / Picklist Values
Candidate	Candidate__c	Master-Detail	Parent Target Connection: Contact (Required)
Job Opening	Job_Opening__c	Master-Detail	Parent Target Connection: Job_Opening__c (Required)
Application Date	Application_Date__c	Date	Standard Date Tracking
AI Candidate Score	AI_Candidate_Score__c	Number	Integer Formatting (Enforced Bounds: 0 to 100)
Offer Amount	Offer_Amount__c	Currency	Standard Currency Ledger Layout (Scale: 18, 2)
Application Status	Application_Status__c	Picklist	Active, Inactive, Offer Submitted
Approval Status	Approval_Status__c	Picklist	Pending, Approved, Rejected

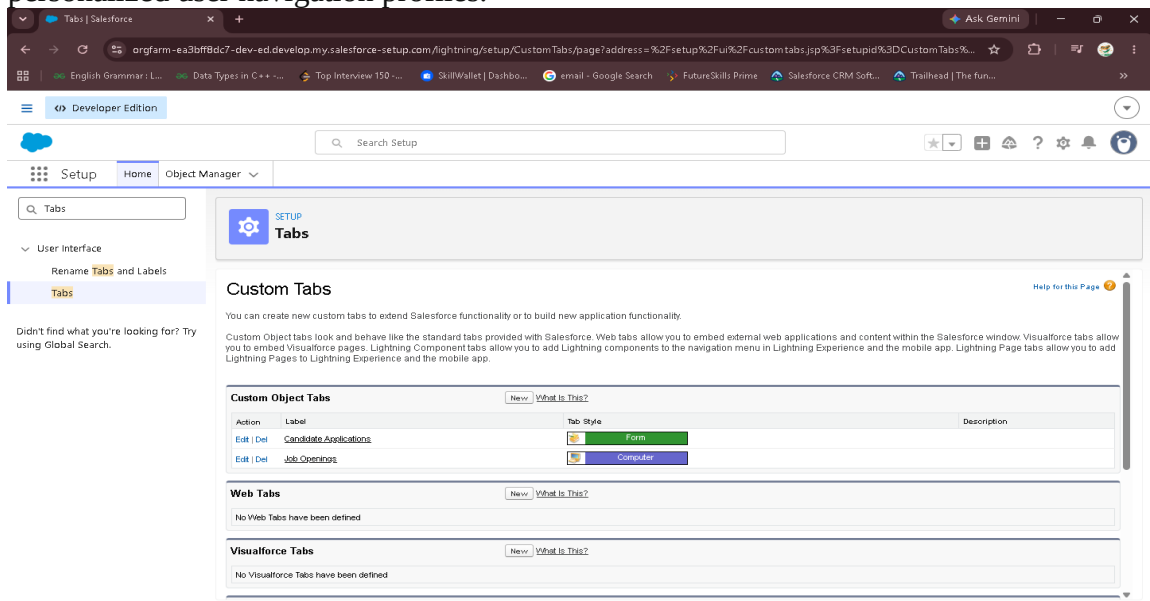


“Figure 4.3: Structural database schema directory for Candidate Application junction table.”

5. User Interface Customization & Workspace Design

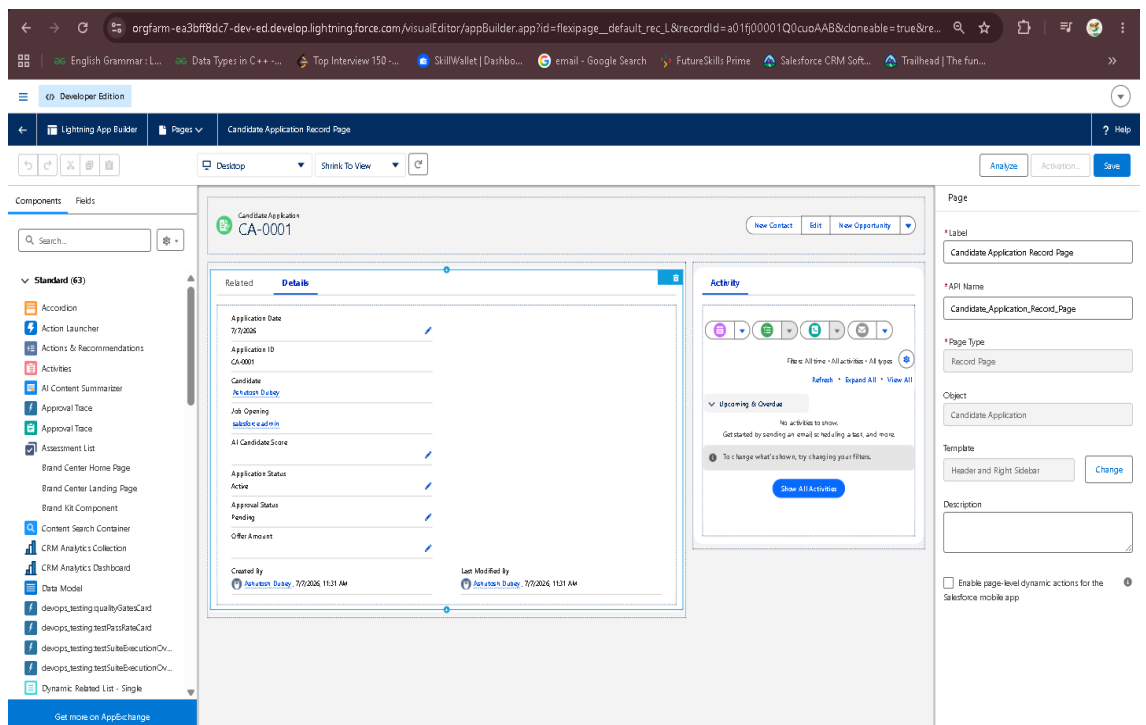
To deliver an intuitive and productivity-driven workspace, user interface components are structured within a custom Lightning console setup:

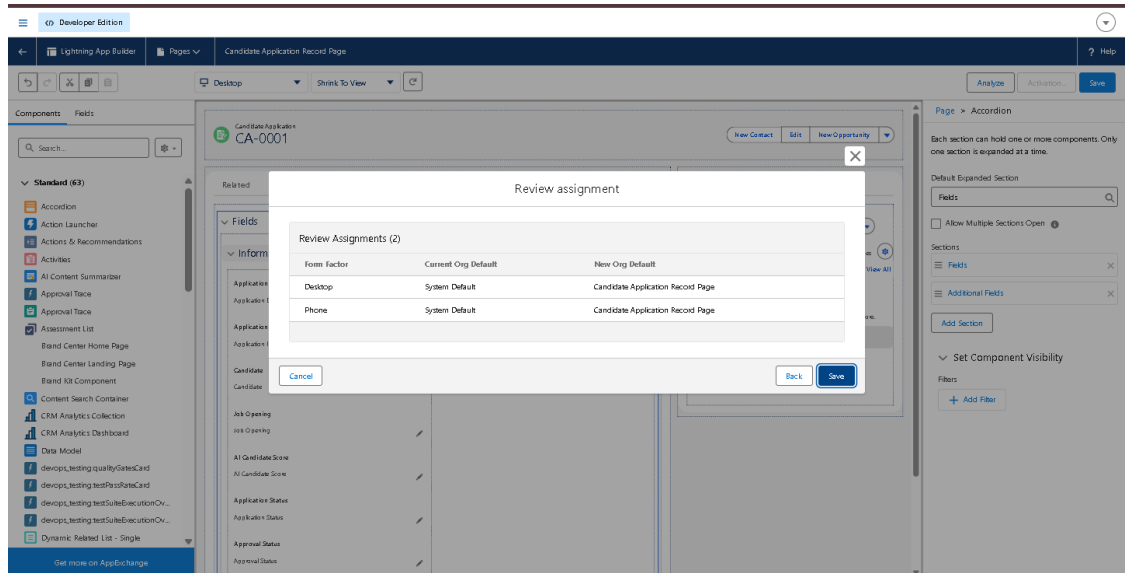
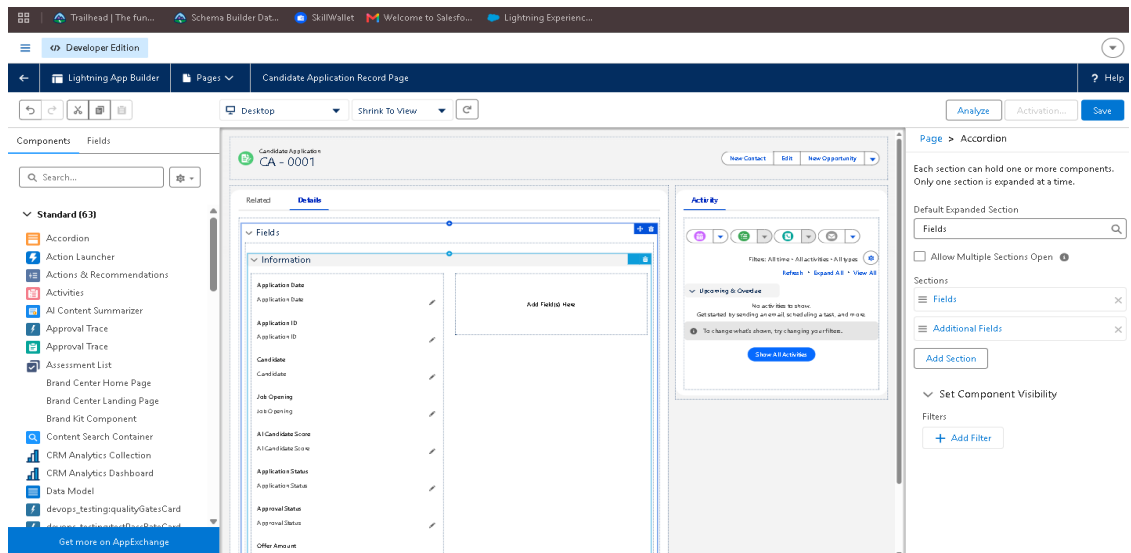
- **Custom Interface Tabs Setup:** Custom object navigation entries are generated to control data access. The **Job Openings** tab uses the `Computer` visual style, and the **Candidate Applications** tab uses the `Form` visual design. Both tabs are set to append automatically to personalized user navigation profiles.



“Figure 5.1: Navigation Custom Object Tabs wizard directory deployment.”

- **Lightning Console Application Configuration:** All structural objects are grouped under a unified Lightning application labeled "**AI-Powered Recruitment & Candidate Management System**". Navigation settings include Candidate Applications, Candidates (Contact), Job Openings, Accounts, Reports, and Dashboards. Access rights are granted exclusively to System Administrator, Recruiter, and Hiring Manager security profiles.
- **Dynamic Forms Architecture Deployment:** Standard record page views are upgraded to **Dynamic Forms** layers. This approach allows administrators to configure field visibility and section arrangements dynamically from the Lightning App Builder without relying on rigid page layouts. This upgrade improves record load performance on both desktop and mobile devices.





“Figure 5.2: Upgrading and activating Lightning Record Pages to cross-platform Dynamic Forms.”

6. Data Integrity Architecture & System Validation Rules

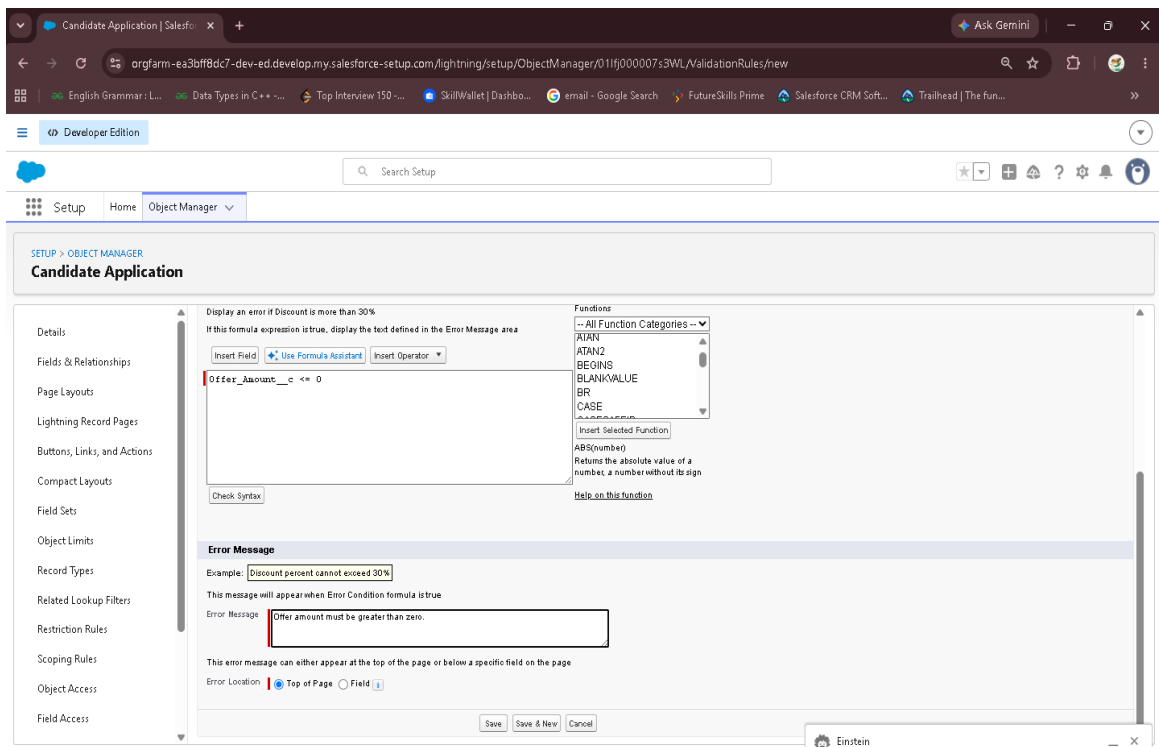
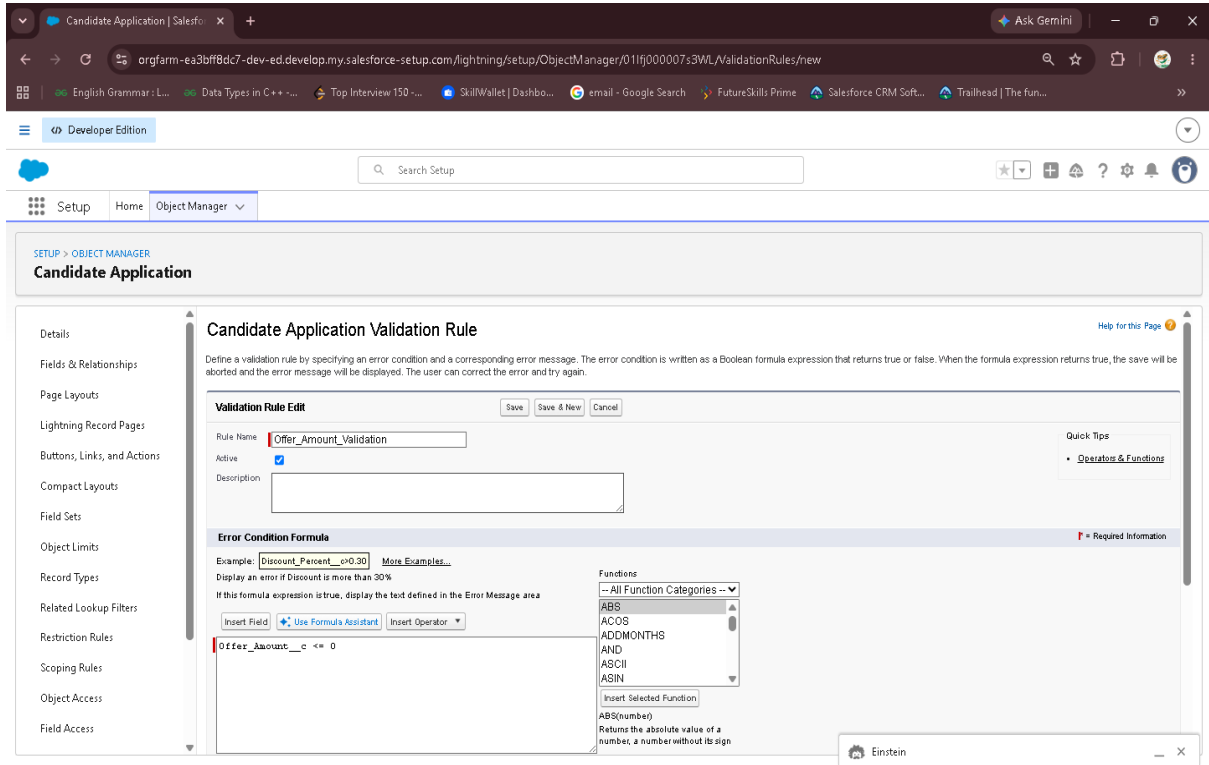
To enforce high data quality and provide clean information for downstream AI workflows, the system uses strict declarative validation formulas. If invalid data values are entered, database saves are blocked, and explicit error labels display to the user.

Rule 1: Offer Amount Validation

- **Target Entity Object:** Candidate Application (Candidate_Application__c)
- **Unique Developer Identifier:** Offer_Amount_Validation
- **Error Condition Evaluation Formula:**

`Offer_Amount__c <= 0`

- **System Presentation Message String:** "Offer amount must be greater than zero."
- **Error Layout Mapping:** Top of page window.



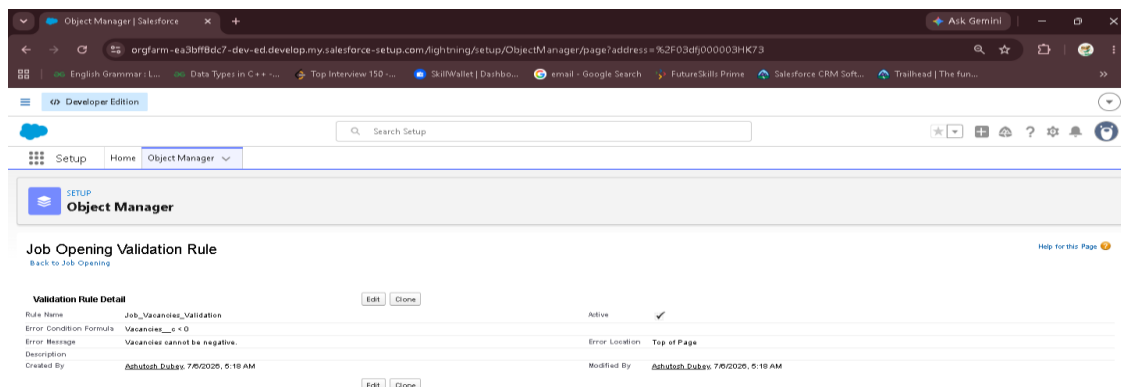
“Figure 6.1: Enforcing positive compensation constraints via Offer Amount validation formula.”

Rule 2: Job Vacancies Validation

- **Target Entity Object:** Job Opening (Job_Opening__c)
- **Unique Developer Identifier:** Job_Vacancies_Validation
- **Error Condition Evaluation Formula:**

```
Vacancies__c < 0
```

- **System Presentation Message String:** "Vacancies cannot be negative."
- **Error Layout Mapping:** Top of page window.



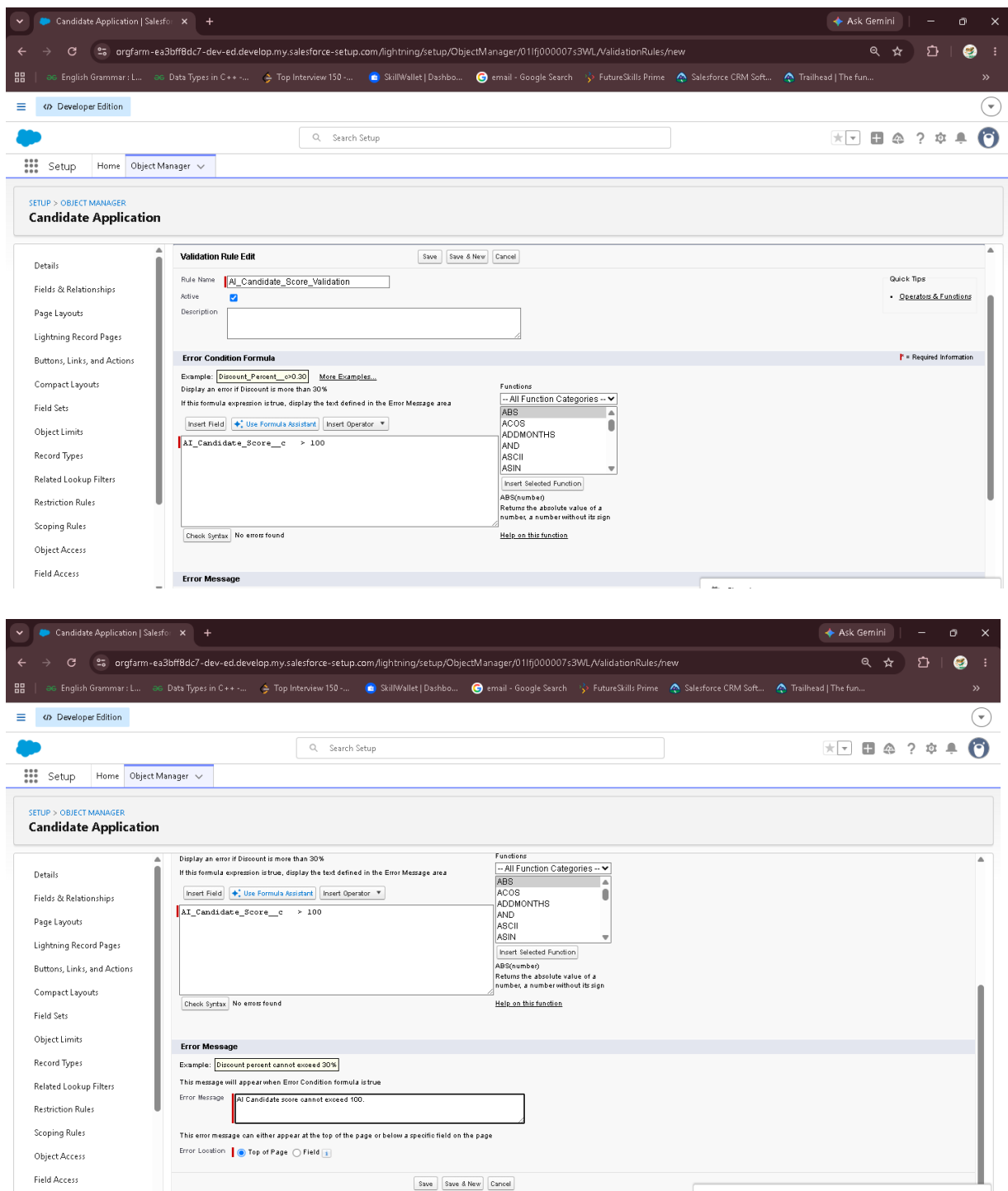
“Figure 6.2: Enforcing vacancy data compliance verified under creator Ashutosh Dubey.”

Rule 3: AI Candidate Score Validation

- **Target Entity Object:** Candidate Application (Candidate_Application__c)
- **Unique Developer Identifier:** AI_Candidate_Score_Validation
- **Error Condition Evaluation Formula:**

```
AI_Candidate_Score__c > 100
```

- **System Presentation Message String:** "AI Candidate score cannot exceed 100."
- **Error Layout Mapping:** Top of page window.



“Figure 6.3: Integrity validation ensuring score matrices do not breach upper limits.”

7. Deduplication & Matching Governance Frameworks

Duplicate candidate records can fragment history tracking and skew downstream AI evaluation scores. The system resolves this vulnerability by establishing a native deduplication engine at the Candidate (Contact) object tier:

- **Custom Matching Rule (Unique Candidate - Email & Phone):** This rule uses a multi-field matching matrix to identify duplicate records. It scans newly submitted candidate entries using **Exact Matching Methods** on the Email and Phone fields. The **Match Blank Fields** parameter is enabled to catch matching profiles across partial data entries.
- **Custom Duplicate Rule (Unique Candidate Email and Phone):** This rule enforces system policies when matching entries are detected. It triggers on both **Record Creation** and **Record Modification**, executing **Allow and Report** behavior. This allows data tracking while logging duplicate attempts for administrative review. When triggered, it presents a clear alert string to the user: "Student Email and Phone Number must be unique."

8. Documented Messaging Architecture (Templates & Alerts)

Automated communication loops use standard Classic plain-text email templates to ensure high delivery rates across corporate firewalls:

8.1 Configuration Template:

`High\_Priority\_Candidate\_Application\_Notification`

Subject Header String: High-Priority Candidate Application Assigned to You

```
Hello {!Candidate_Application__c.OwnerFullName}, A candidate application has been identified as high priority by AI and requires immediate attention. Candidate Application Details: Candidate Name: {!Candidate_Application__c.Candidate__r.Name} Job Opening: {!Candidate_Application__c.Job_Opening__r.Name} Experience Level: {!Candidate_Application__c.Candidate__r.Experience_Level__c} AI Candidate Score: {!Candidate_Application__c.AI_Candidate_Score__c} Application Date: {!Candidate_Application__c.Application_Date__c} Please contact the candidate and proceed with the recruitment process within 24 hours. Thank you, AI Recruitment Management System
```

8.2 Configuration Template:

`High\_Salary\_Candidate\_Offer\_Approval\_Notification`

Subject Header String: Candidate Offer Awaiting Salary Approval

```
Dear {!Candidate_Application__c.OwnerFullName}, A candidate job offer with a high salary has been submitted for your approval. Please review the candidate application details below: Candidate Offer Details: Candidate Name: {!Candidate_Application__c.Candidate__r.Name} Job Opening: {!Candidate_Application__c.Job_Opening__r.Name} Experience Level: {!Candidate_Application__c.Candidate__r.Experience_Level__c} Proposed Offer Amount: {!Candidate_Application__c.Offer_Amount__c} AI Candidate Score: {!Candidate_Application__c.AI_Candidate_Score__c} Application Status: {!Candidate_Application__c.Application_Status__c} Please take the necessary action at your earliest convenience. Thank you, AI Recruitment Management System
```

8.3 Configuration Template: `Candidate\_Application\_Approval\_Status\_Notification`

Subject Header String: Candidate Offer Approval Decision

```
Hello {!Candidate_Application__c.OwnerFullName}, The approval process for a candidate offer has been completed. Application Summary: Candidate Name: {!Candidate_Application__c.Candidate__r.Name} Job Opening: {!Candidate_Application__c.Job_Opening__r.Name} Approval Status: {!Candidate_Application__c.Approval_Status__c} Final Offer Amount: {!Candidate_Application__c.Offer_Amount__c} Please proceed with the next steps in the recruitment process based on the approval outcome. Thank you, AI Recruitment Management System
```

8.4 System Email Alerts Execution Mapping

- **Alert 1 (High Priority Ingestion Notification):** Built on Candidate_Application__c. Tied to template 8.1. Routes automatically to the target user field: Candidate Application Owner.
- **Alert 2 (High Salary Offer Submitted Alert):** Built on Candidate_Application__c. Tied to template 8.2. Routes automatically to the target user field: Manager.
- **Alert 3 (Approval Sequence Outcome Status Alert):** Built on Candidate_Application__c. Tied to template 8.3. Routes automatically to the target user field: Candidate Application Owner.

The screenshot displays the Salesforce Setup interface. The top navigation bar includes the 'Developer Edition' label and a search bar labeled 'Search Setup'. Below this, the 'Setup' tab is active, showing a sidebar with 'Email Alerts' selected. The main content area is titled 'Email Alerts' and contains a table listing three email alerts. The table has columns for 'Action', 'Description', 'Email Template Name', 'Object', and 'Last Modified Date'. The first alert is 'Candidate Application Approval Status Email Alert' for the 'Candidate Application' object, last modified on 7/8/2026. The second is 'High Priority Candidate Application Email Alert' for the 'Candidate Application' object, also last modified on 7/8/2026. The third is 'High Salary Candidate Offer Approval Email Alert' for the 'Candidate Application' object, last modified on 7/8/2026. A 'New Email Alert' button is located above the table. The interface also includes a 'Help for this Page' link and a 'Global Search' prompt in the sidebar.

Action	Description *	Email Template Name	Object	Last Modified Date
Edit Del	Candidate Application Approval Status Email Alert	Candidate Application Approval Status Notification	Candidate Application	7/8/2026
Edit Del	High Priority Candidate Application Email Alert	High Priority Candidate Application Notification	Candidate Application	7/8/2026
Edit Del	High Salary Candidate Offer Approval Email Alert	High Salary Candidate Offer Approval Notification	Candidate Application	7/8/2026

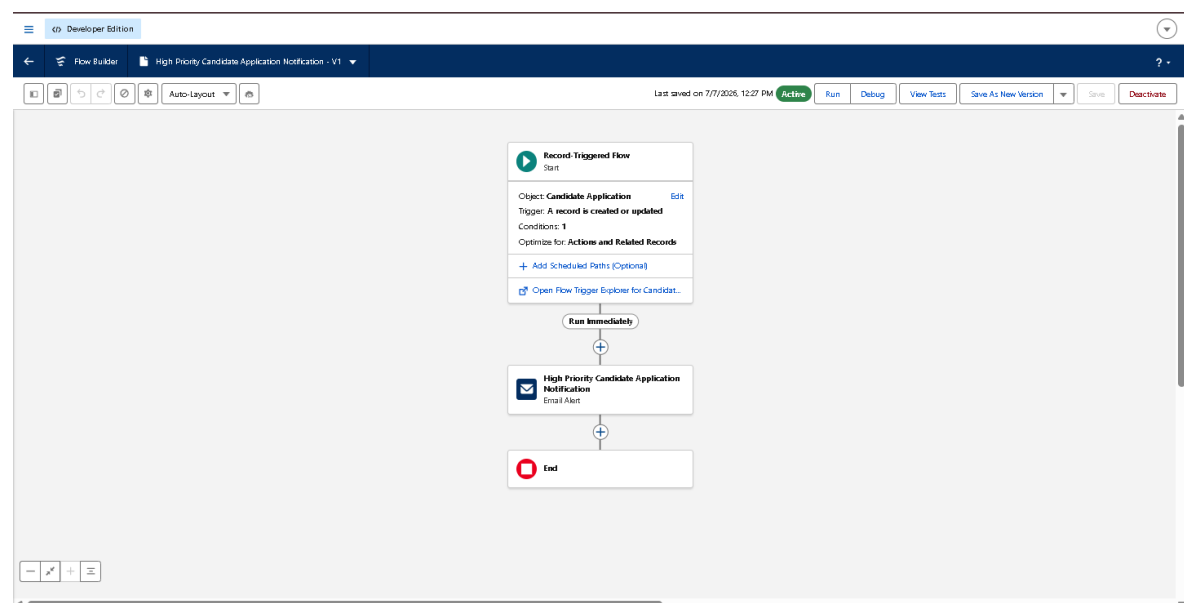
“Figure 8.1: Unified automated system Email Alerts registry catalog.”

9. Declarative Automation Engine (Flow Builder Architecture)

The system replaces manual milestone tracking with four declarative business process automation paths configured in Flow Builder:

Flow 1: High Priority Candidate Application Notification (Record-Triggered)

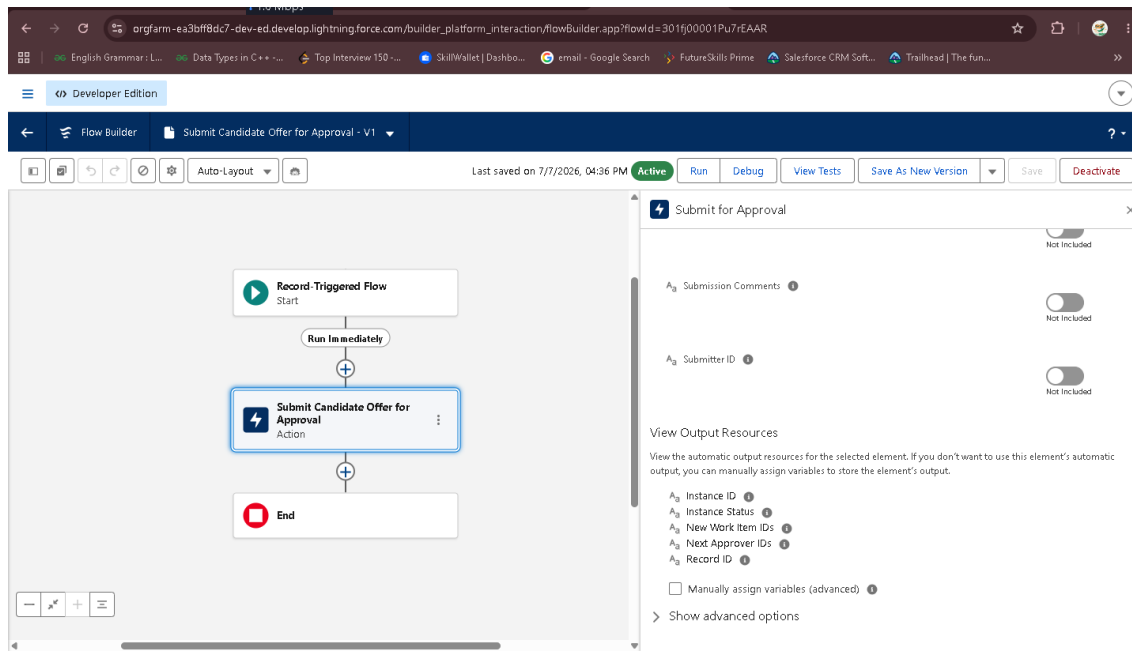
Triggers when a `Candidate_Application__c` record is created or updated. It filters operations using the entry criteria formula: `AI_Candidate_Score__c >= 85`. The flow is set to execute ***every time a record is modified to meet the condition requirements***, optimized for ****Actions and Related Records****. When triggered, it calls the `High Priority Candidate Application Email Alert` action node, mapping the current record identity token (`{!$Record.Id}`) to pass application details dynamically.



“Figure 9.1: Building and syntax-checking the record-triggered high priority notification flow.”

Flow 2: High Salary Offer Submission for Approval (Record-Triggered)

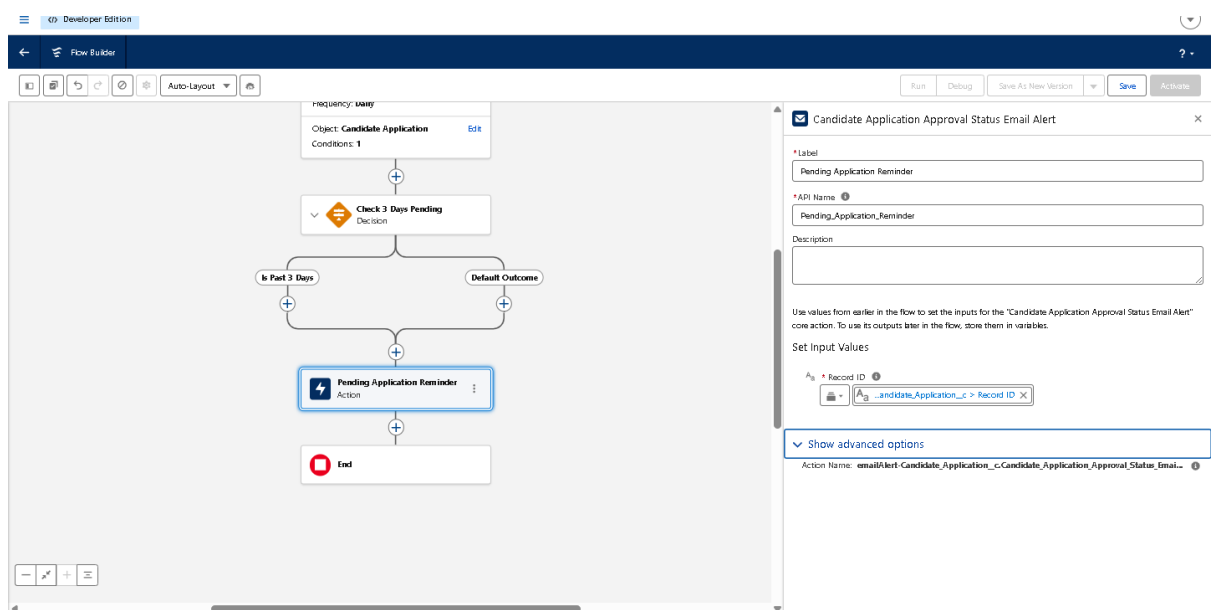
Triggers when a `Candidate_Application__c` record is created or updated. It filters operations using the entry criteria formula: `Offer_Amount__c > 1000000`. Optimized for ****Actions and Related Records****, it triggers a core platform action node labeled ****Submit for Approval****. This action targets the `High Salary Candidate Offer Approval` process template, automatically locking the record data row from further changes and routing it to the designated reviewer.

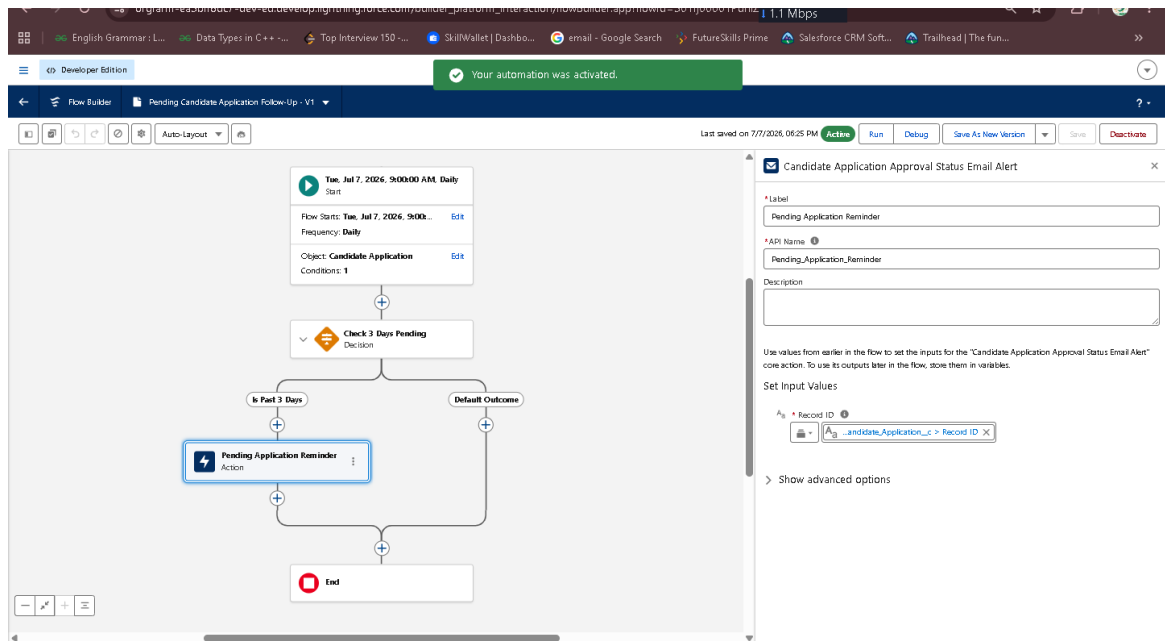
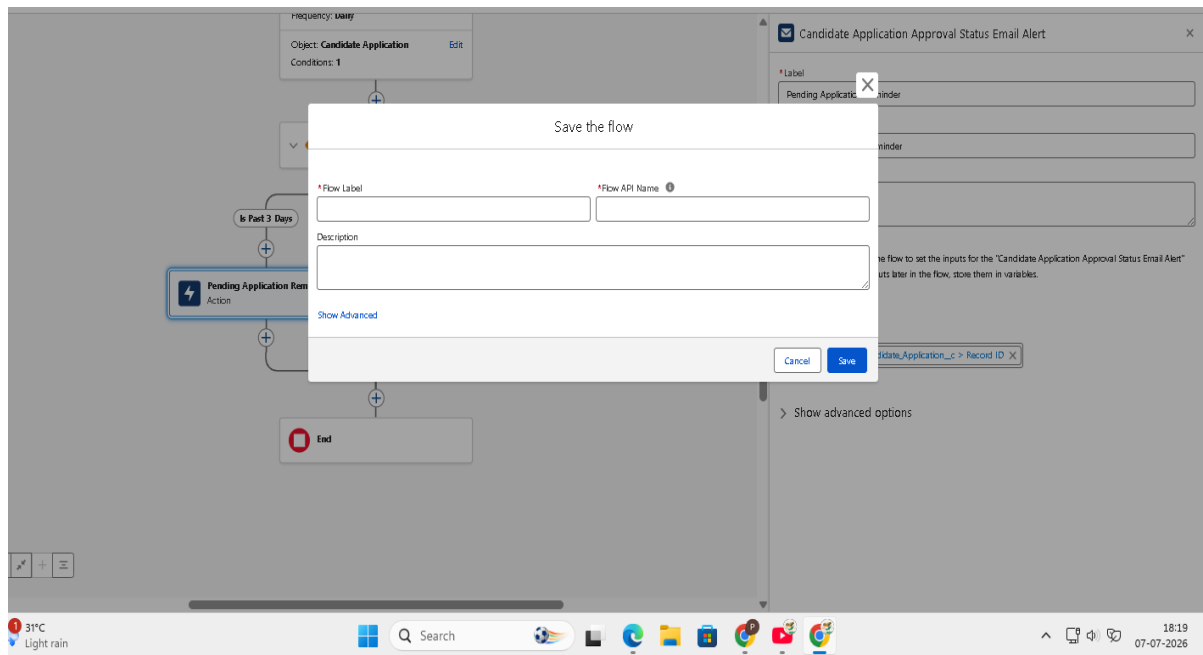


“Figure 9.2: Activated operational canvas for the automated approval submission workflow flow.”

Flow 3: Pending Candidate Application Follow-Up (Scheduled-Triggered)

Runs on an automated daily schedule at a designated morning interval (e.g., 9:00 AM) to evaluate all candidate records against two criteria conditions: `Approval_Status__c == 'Pending' AND LastModifiedDate < TODAY() - 3`. When records match this criteria loop, the flow calls an email alert action node labeled ****Pending Application Reminder****. This system automatically sends follow-up notifications to the reviewer, preventing candidates from getting stuck in the pipeline.

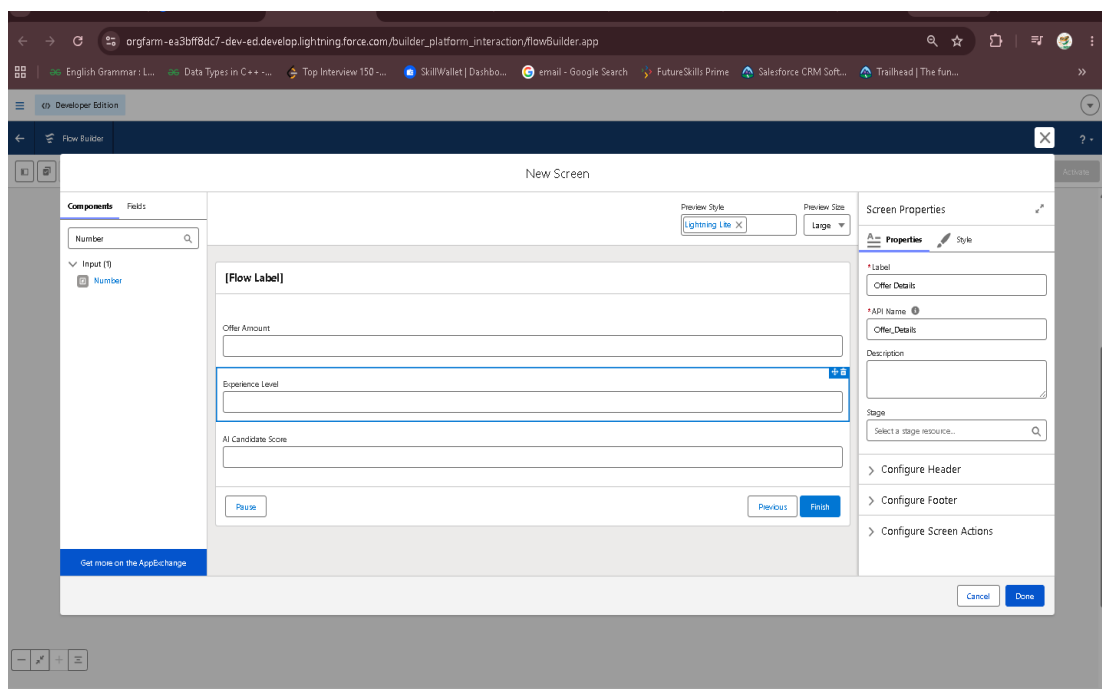
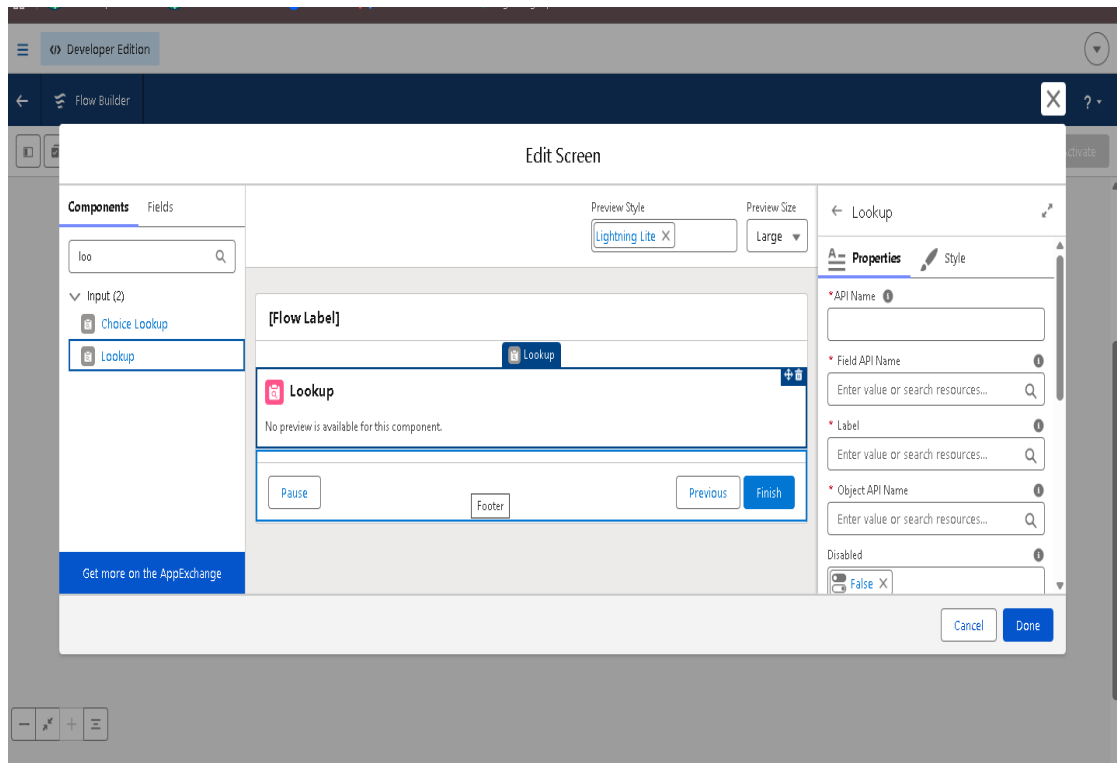




“Figure 9.3: Live production-ready scheduled batch automation engine deployment canvas.

Flow 4: Candidate Application Creation Wizard (Interactive Screen Flow)

Provides recruiters with a guided multi-screen interface to register applications uniformly. ****Screen 1 (Candidate Details)**** captures parent lookup selections for Candidate and Job Opening along with the Application Date. ****Screen 2 (Offer Details)**** captures manual inputs for proposed Offer Amount, Experience Level, and baseline AI Candidate Score fields. A background assignment calculation step validates financial parameters before routing variables to a `Create Records` node to insert the target `Candidate_Application__c` row. Upon successful insertion, ****Screen 3 (Confirmation)**** displays validation text: "Candidate Application Created Successfully". (API Identifier: `Candidate_Application_Screen_Flow`).



“Figure 9.4: Final interactive UI fields layout block mapping inside Screen Flow wizard component.”

10. Enterprise Approval Routing & Policy Governance

To protect institutional budgets, the system locks high-value compensation offers until they are reviewed and approved by an authorized manager. This structural governance track is handled by an active, multi-action Classic Approval Process definition:

Process Schema Blueprint: `High Salary Candidate Offer Approval`

- **Target Base Object Binding:** Candidate Application (Candidate_Application__c)
- **Process Entry Filtering Logic Matrix:**

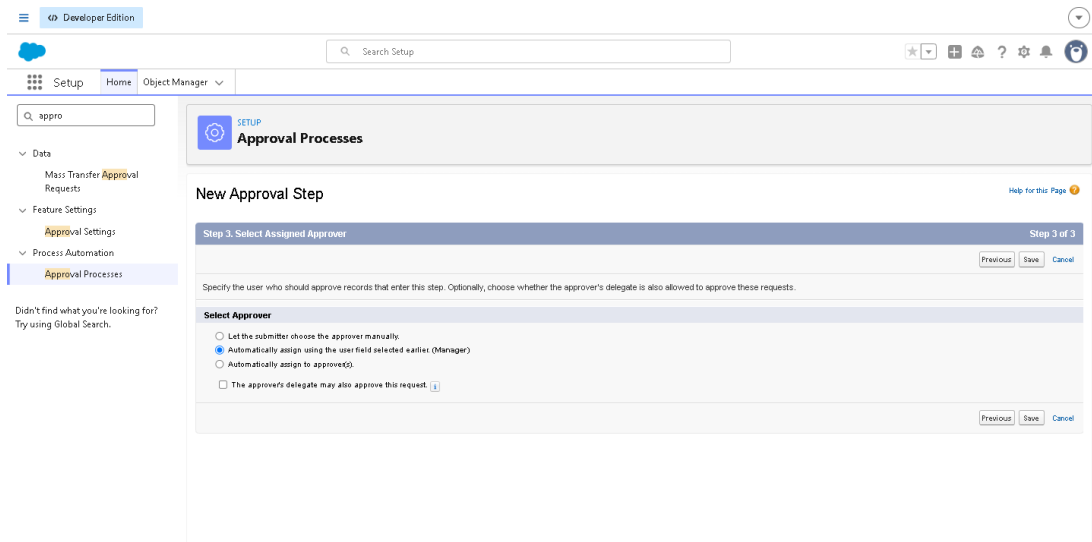
```
(Candidate_Application__c.Offer_Amount__c > 1000000) AND  
(Candidate_Application__c.Application_Status__c == 'Offer  
Submitted')
```

The screenshot displays the Salesforce Classic Approval Process definition interface. A 'Lookup' dialog box is open, showing a list of values to select. The main interface is at 'Step 2 of 6' for defining entry criteria. It includes a table for specifying criteria with columns for Field, Operator, and Value. The criteria defined are: 'Candidate Application: Offer Amount' greater than '1000000' AND 'Candidate Application: Application Status' equals 'Offer Submitted'. A red error message is visible: 'Error: Picklist value "Offer Submitted" does not exist for "Application Status" field.' Navigation buttons 'Previous', 'Next', and 'Cancel' are at the bottom.

Field	Operator	Value
Candidate Application: Offer Amount	greater than	1000000
Candidate Application: Application Status	equals	Offer Submitted
--None--	--None--	
--None--	--None--	
--None--	--None--	

“Figure 10.1: Defining defensive budgetary entry thresholds for compensation approval.”

- **Record Data Lock Policy:** Records are locked during evaluation; only System Administrators retain manual edit permissions.
- **Approver Routing Rules:** Automatically routes records based on the organization's user relationship schema, targeting the ****Manager**** listed on the submitter's User record.



“Figure 10.2: Enforcing organizational matrix reporting directly to the owner's Manager.

- **Assigned Template Mapping:** High Salary Candidate Offer Approval Notification
- **Approval Page Layout Display:** Candidate Name, Job Opening, Offer Amount, AI Candidate Score, Application Status.

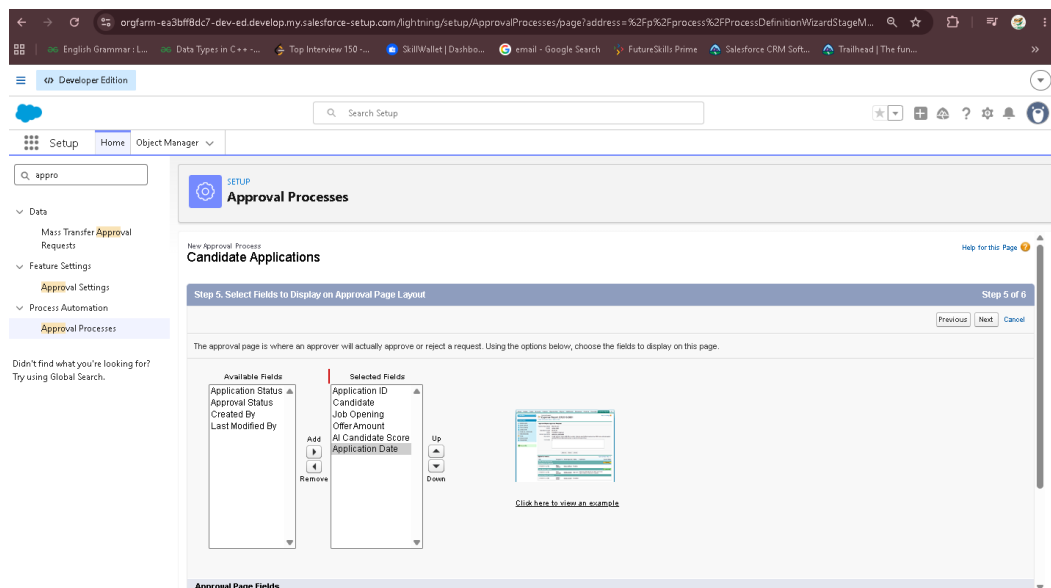


Figure 10.3: Layout properties configuration for the manager approval presentation pane.”

- **Authorized Submitters Ledger:** Restructured to target the ****Record Owner****, allowing frontline recruiters to launch approval routing directly from the interface layout.

Configured Sequence Process Actions

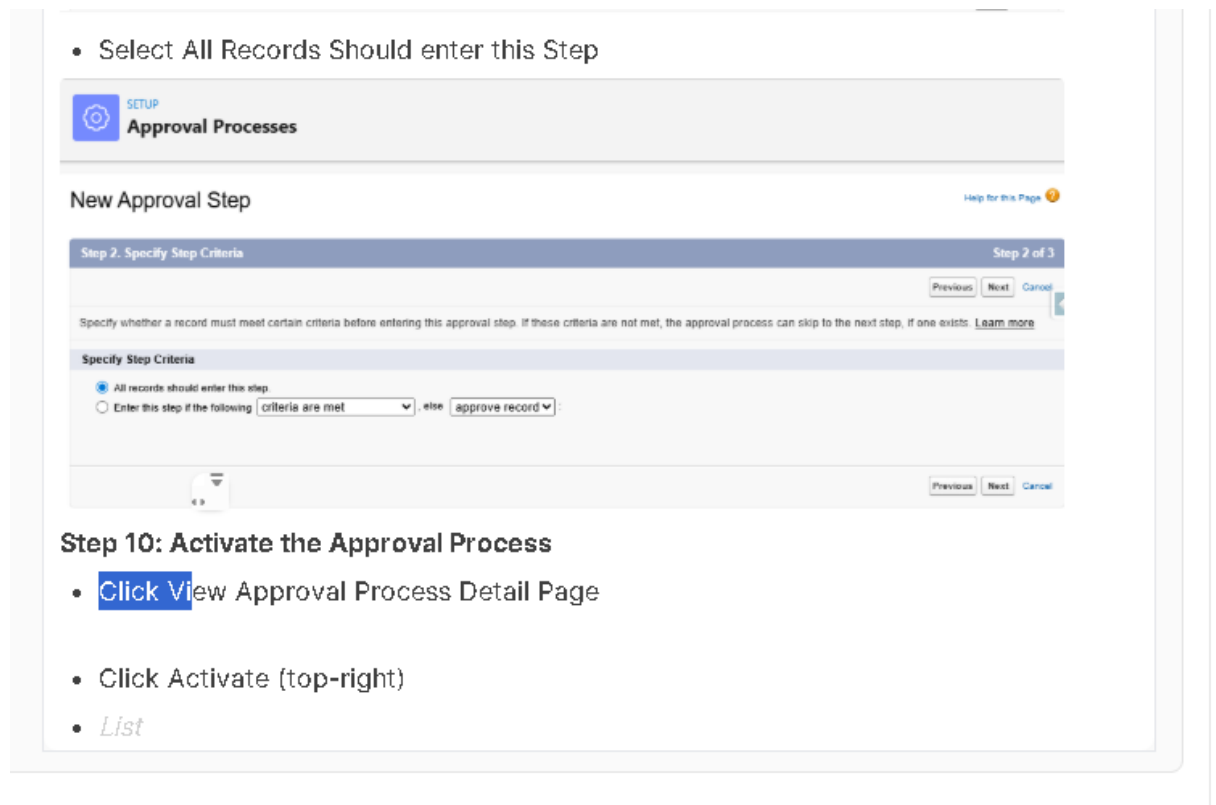
- **Initial Submission Actions:** Locks record access and fires Alert 2 to notify the manager.
- **Final Approval Step Actions (Field Update):** Triggers automatically upon managerial sign-off.

Runs a field update that changes the `Approval_Status__c` picklist value to **Approved**.

- **Final Rejection Step Actions (Field Update):** Triggers if a manager denies the compensation plan. Runs a field update that changes the `Approval_Status__c` picklist value to **Rejected**.

The screenshot displays the Salesforce Setup interface for configuring Field Updates. The page title is "Field Updates". The left sidebar shows navigation options: "Data", "Feature Settings", and "Process Automation". The main content area is titled "Field Update Edit" and includes a search bar. The "Identification" section contains fields for "Name" (Set Approval Status to F), "Unique Name" (Set_Approval_Status__t), "Description", "Object", "Field to Update" (Approval Status), "Field Data Type" (Picklist), and "Re-evaluate Workflow Rules after Field Change" (checkbox). The "Specify New Field Value" section shows "Picklist Options" with three radio buttons: "The value above the current one", "The value below the current one", and "A specific value". The "A specific value" option is selected, and the value "Rejected" is entered in the adjacent dropdown menu.

“Figure 10.4: Constructing automated Field Updates to lock status changes upon rejection.”



“Figure 10.5: Executing full process path structure deployment and final activation loops.”

11. Agentforce AI Integration & Trust Guardrails

Rather than conversational chat windows, the system integrates generative AI models directly into standard record pages via Agentforce and Prompt Builder templates. This provides data insights directly to hiring managers within their daily workspace.

11.1 System Trust Guardrails & Governance Controls

To prevent data leaks and protect candidate privacy during external large language model (LLM) evaluations, the platform uses native **AI Trust & Permissions** controls:

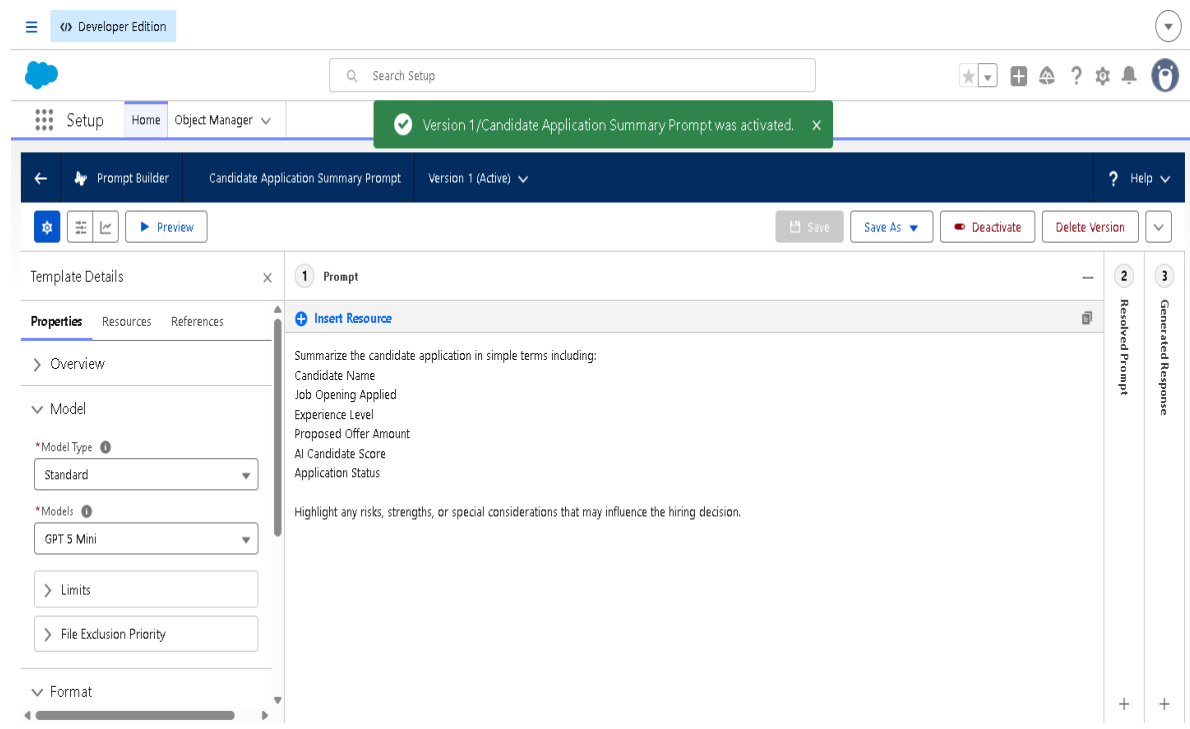
- **Data Masking:** Automatically scrambles personal candidate communication parameters before information passes out of the organizational boundary.
- **Zero Data Retention Policy:** Enforces rules ensuring no prompt strings or context logs are cached or used for external model training.
- **System Audit Logging:** Tracks all prompt template runs and AI generations for administrative security review.

- **Access Profiles:** Restricted to authorized System Administrator and Hiring Manager profiles.

11.2 Prompt Template Registry Definitions

Template 1: Candidate Application Summary Prompt (Summary Type)
***Object Binding:** Candidate_Application_c | ***API Name:** Candidate_Application_Summary_Prompt
Context Instructions Matrix:

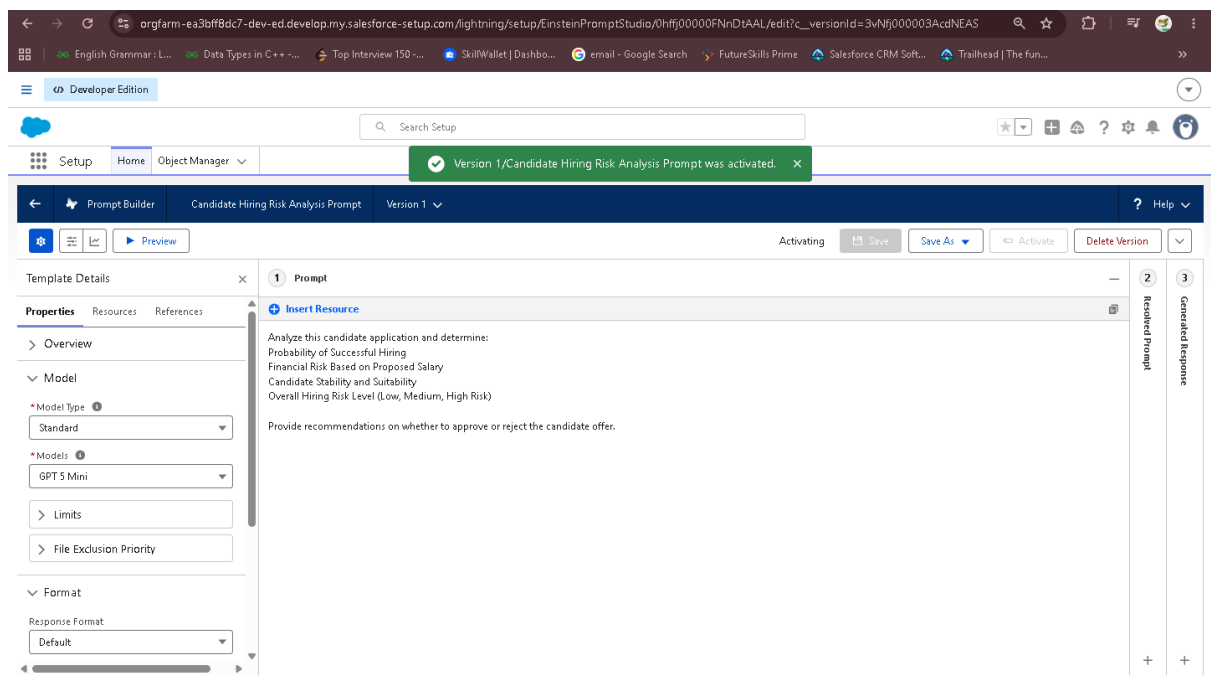
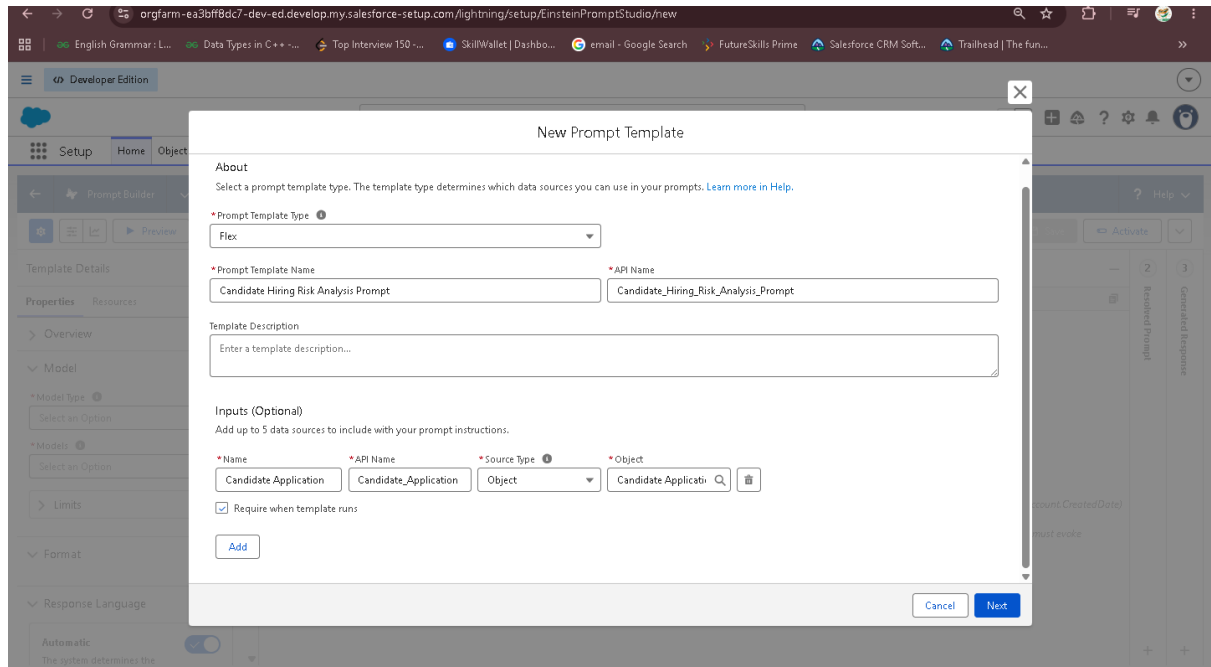
Summarize the candidate application in simple terms including: -
 Candidate Name - Job Opening Applied - Experience Level - Proposed
 Offer Amount - AI Candidate Score - Application Status Highlight any
 risks, strengths, or special considerations that may influence the
 hiring decision.



“Figure 11.1: Active GPT 5 Mini generative AI template Studio window for summary indexing.”

Template 2: Candidate Hiring Risk Analysis Prompt (Flex Type)
***Object Binding:** Candidate_Application_c | ***API Name:** Candidate_Hiring_Risk_Analysis_Prompt
Context Instructions Matrix:

Analyze this candidate application and determine: 1. Probability of Successful Hiring 2. Financial Risk Based on Proposed Salary 3. Candidate Stability and Suitability 4. Overall Hiring Risk Level (Low, Medium, High Risk) Provide recommendations on whether to approve or reject the candidate offer.

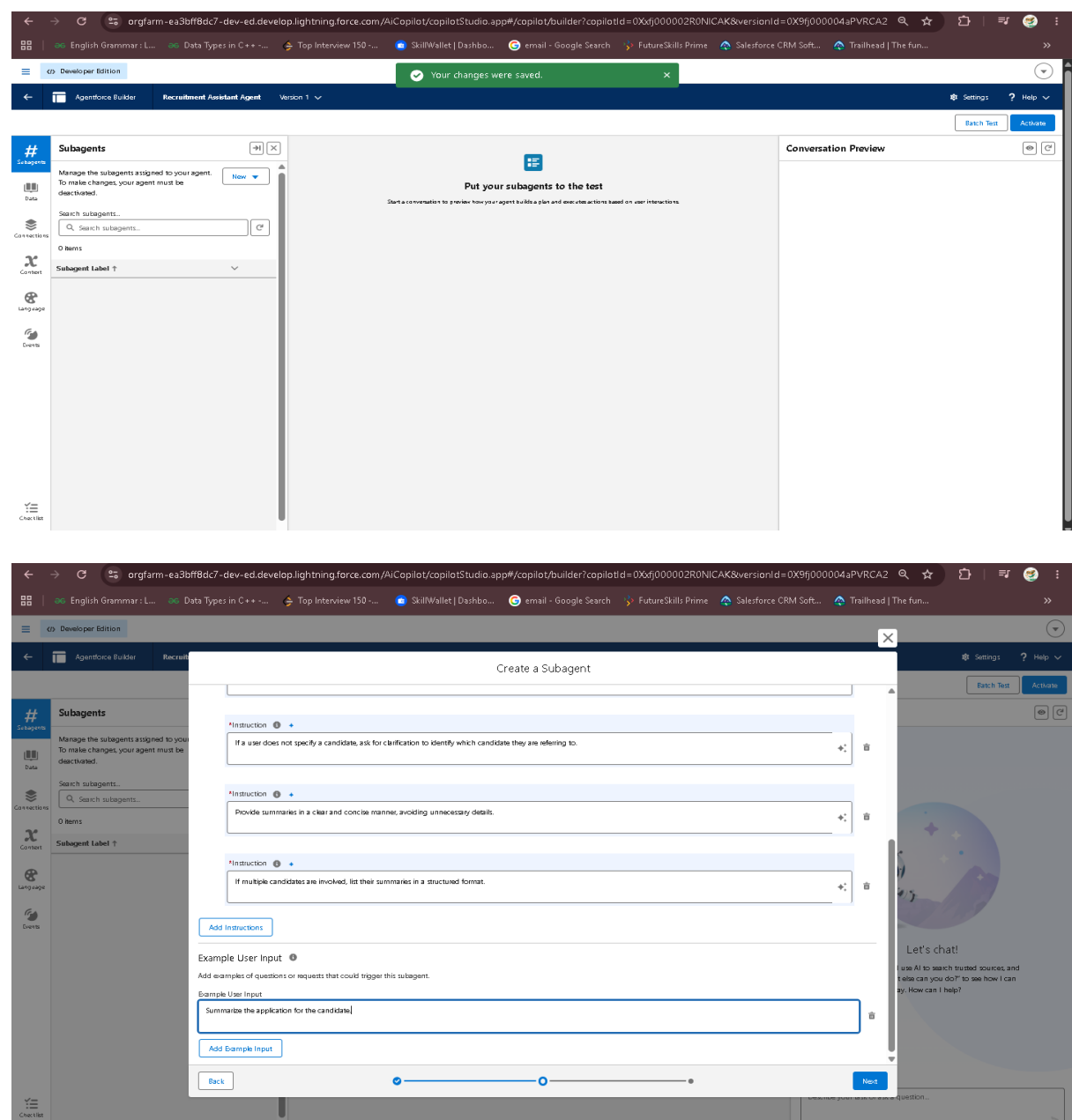


“Figure 11.2: Structuring object input bounds and activating the autonomous risk identification prompt matrix.”

11.3 Agent Deployment Architecture

The system packages these topics into an autonomous runtime engine labeled the **Recruitment Assistant Agent**. Inside the agent actions setup, the prompt templates are linked directly to operational targets: **Action: Generate Candidate Application Summary** calls Template 1, and **Action: Analyze Hiring Risk** calls Template 2.

The ****Agent Data Library**** grants the assistant secure, read-only access to Candidate (`Contact`), `Job\_Opening\_\_c`, and `Candidate\_Application\_\_c` data records. The agent automatically respects native field-level security settings based on the running user profile, ensuring users only see information they are authorized to access.



“Figure 11.3: Deploying specialized, autonomous subagent instructions inside the core Agentforce Builder framework.”

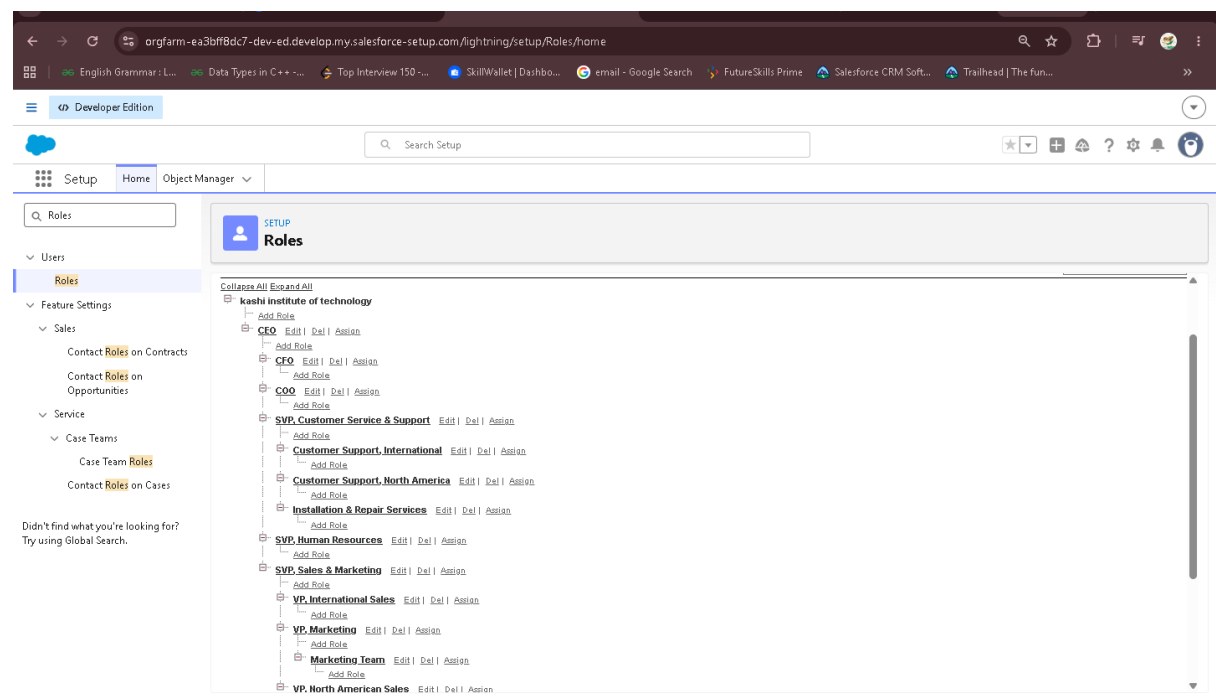
12. Security Architecture, Profiles, & Access Control

Data access boundaries are established using a structured role hierarchy combined with custom profile permission matrices, preventing unauthorized access across corporate tiers.

12.1 Organizational Visibility Hierarchy

The organization's hierarchy controls record visibility based on user roles:

Hiring Manager Role: Maintained directly beneath the CEO tier. This role grants full visibility across all underlying recruiter pipelines and candidate records. **Recruiter Role:** Maintained directly beneath the Hiring Manager role. This structure restricts visibility so frontline recruiters focus on their assigned openings and candidates.



“Figure 12.1: Mapping user access bounds via the active consolidated corporate Role Hierarchy tree directory.”

12.2 Profile Permission Security Configurations

Recruiter Profile Permissions Matrix

- **Custom App Default:** Automatically loads the `AI-Powered Recruitment & Candidate Management System` console application interface.
- **Object Permission Access Layer:**
 - Standard Contact (Candidates): Read, Create, Edit (Delete: Blocked)
 - Job Opening (Job_Opening__c): Read, Create, Edit (Delete: Blocked)

- **Candidate Application** (Candidate_Application__c): Read, Create, Edit (Delete: Blocked)
- **Standard Reports and Dashboards:** Read-Only Access
- **AI Capabilities Boundary:** Granted read-only visibility into the AI_Candidate_Score__c metric field. This profile is blocked from editing prompt frameworks or manually bypassing approval rules.
- **Session Policies:** Enforces automatic logouts after 2 hours of user inactivity. Passwords expire every 90 days with a minimum length of 8 characters.

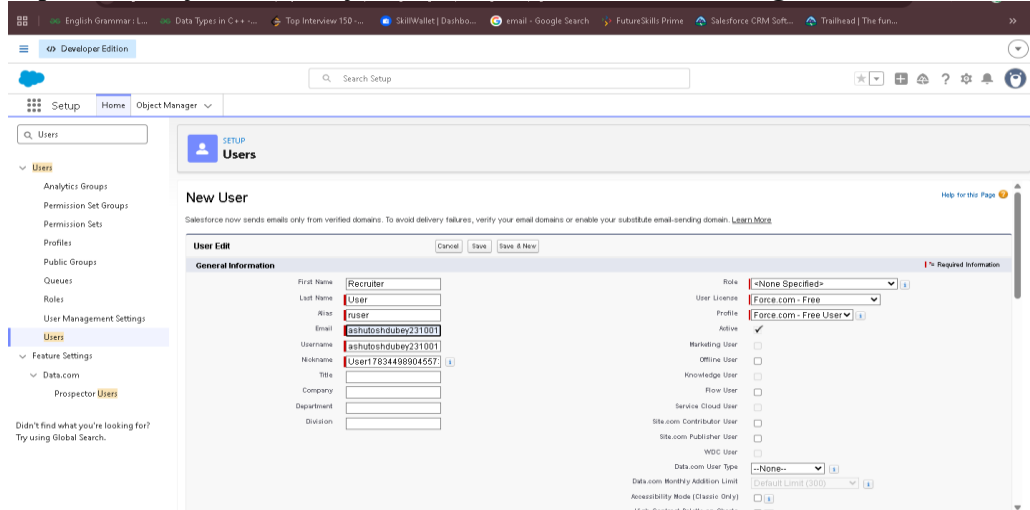


Figure 12.2: Administrative provisioning setup screen for ingesting a new Recruiter User into the platform.”

Hiring Manager Profile Permissions Matrix

- **Custom App Default:** Automatically loads the `AI-Powered Recruitment & Candidate Management System` console application interface.
- **Object Permission Access Layer:** Full Read, Create, Edit, and Delete permissions across all custom tracking objects, standard candidate contacts, reports, and team dashboard grids.
- **Policy Control Layer:** Full authority to approve or reject high-salary offers, modify prompt configurations, change model choices, and manually override AI evaluations.
- **Session Policies:** Enforces automatic logouts after 2 hours of user inactivity.

13. Data Migration & Legacy Ingestion Strategy

To replace legacy systems and spreadsheets cleanly, migration activities are split across specific tools to maintain relationship linkages:

- **Candidate Data Migration Ingestion:** Historic candidate profiles are imported into the standard `Contact` table via the ****Data Import Wizard****, matching unique records against name and email fields.

- **Job Openings & Applications Ingestion:** Bulk historical data loads are run through the ****Data Loader Engine**** tool to efficiently map data volumes into custom `Job_Opening__c` and `Candidate_Application__c` structures.
- **Relational Mapping Validation:** Parent record IDs are verified during data mapping to ensure candidate applications reconnect correctly to their corresponding parent contact entries and job opening records.

14. System Verification & Quality Testing Logs

System configurations were verified using clear test case execution scripts inside a sandbox environment prior to final deployment:

ID	Test Scenario Description	Execution Input / Parameter Context	Expected Outcome Vector	Status
TC-01	Verify negative vacancy validation logic.	Set <code>Vacancies__c</code> = -3 on Job Opening record.	System blocks save; displays message: "Vacancies cannot be negative."	PASS
TC-02	Verify zero-value offer validation logic.	Set <code>Offer_Amount__c</code> = 0 on Candidate App record.	System blocks save; displays message: "Offer amount must be greater than zero."	PASS
TC-03	Verify out-of-bounds AI score limits.	Set <code>AI_Candidate_Score__c</code> = 120 on record save.	System blocks save; displays message: "AI Candidate score cannot exceed 100."	PASS
TC-04	Verify high-priority flow automation trigger.	Insert Candidate Application with an AI Score of 90.	Record-triggered flow executes immediately; fires email alert to owner.	PASS

ID	Test Scenario Description	Execution Input / Parameter Context	Expected Outcome Vector	Status
TC-05	Verify high-salary approval routing process.	Update Candidate Application with Offer = 1,500,000.	Flow intercepts save, locks application data, and routes to Manager.	PASS
TC-06	Verify data deduplication matching rule logic.	Insert Contact duplicate containing identical Email/Phone.	Duplicate rule blocks data ingestion and displays system alert text.	PASS
TC-07	Verify interactive creation screen flow.	Run Candidate_Application_Screen_Flow wizard layout.	Multi-screen inputs collect, process, and insert the record correctly.	PASS

15. Change Management, Support, & Platform Maintenance

To drive user adoption and maintain long-term system stability post-launch, specific change management steps are implemented:

- **Change Management & Adoption Strategy:** Agentforce chat interfaces and AI prompt summary generation blocks are embedded directly into standard Candidate record layout fields. This approach minimizes complexity by keeping insights inside the recruiter's daily workspace, requiring minimal user training.
- **System Monitoring Protocols:** System administrators review the ****Failed Flow Interviews**** log dashboard weekly to catch unexpected errors early. They also run periodic checks on prompt executions to evaluate accuracy, adjust instruction matrices, and prevent model drift.
- **Seasonal Update Management:** Testing protocols require running full regression evaluations across all automated flows and validation rules inside a sandbox environment before major Salesforce seasonal releases go live.

16. Strategic Platform Advantages & Future Scope

Realized Platform Advantages

- **Technical Debt Reduction:** Built completely with declarative tools, eliminating the development and maintenance costs of custom Apex code.
- **Pipeline Velocity Optimization:** Replaces disjointed tracking tools with automated record routing, reducing candidate drop-out rates.
- **Budget Compliance Control:** High-salary validation caps are enforced at the database level, preventing unauthorized compensation offers.
- **Accelerated Profile Screenings:** Grounded prompt templates summarize applicant records instantly, reducing manual evaluation time for hiring teams.

Documented Future Scope

- **Automated Interview Scheduling Integration:** Expand declarative flow paths to connect to external calendar applications for automated interview coordination.
- **Public Candidate Chat Portals:** Extend internal prompt capabilities into a public-facing Agentforce portal, allowing external candidates to track their application status directly.
- **Advanced Analytics Pipelines:** Build aggregate historical tracking dashboards to trace time-to-hire speeds and recruiter conversion metrics across different departments.

17. Conclusion & Reference Catalog

The successful deployment of the AI-Powered Job Recruitment & Candidate Management System demonstrates how organizations can use Salesforce's declarative framework to build robust enterprise applications. By connecting custom object relationships, automated validation steps, multi-tier approval paths, and Agentforce AI architectures, the system standardizes candidate tracking and protects data integrity without the overhead of custom programming.

Reference Catalog

- *Salesforce Platform Security Controls & User Permission Configurations Manual* (help.salesforce.com)
- *Salesforce Trailhead Frameworks: Declarative Automation Design & Flow Builder Best Practices* (trailhead.salesforce.com)
- *Agentforce AI Implementation Guidelines & Prompt Engineering Design Architecture Standards* (developer.salesforce.com)
- *SmartBridge Virtual Internship Program Track Documentation: Advanced Salesforce Architecture & Generative AI Enterprise Ingestion Frameworks.*